

ANISH GOEL

📍 Jagraon, Punjab 142026

☎ +91-9814220203 ✉ anishgoel1129@gmail.com 🔗 [linkedin.com/in/anishgoel1](https://www.linkedin.com/in/anishgoel1) 🌐 anishgoel.vercel.app

PROFESSIONAL SUMMARY

Mechatronics Engineering undergraduate specializing in autonomous systems, embedded electronics, and robotics. Experienced in end-to-end design and fabrication across marine, aerial, and mobile platforms.

EDUCATION

Thapar Institute of Engineering and Technology

Aug 2023 – May 2027

Bachelor of Engineering in Mechatronics (CGPA: 9.6)

Patiala, Punjab

PROJECTS

Autonomous Underwater Vehicle (AUV)

Patent No. 202611022081

- Built a 1.5 m, 25 kg underactuated AUV using single-thruster propulsion and fin actuation, slashing hydrodynamic drag by 22%.
- Engineered a modular internal chassis securing a 5 kg payload capacity.

Autonomous Surface Vehicle (ASV)

Patent No. 202611063313

- Built a waterproof modular chassis in CAD and deployed ROS navigation nodes for river mapping.
- Cut manual surveying overhead by integrating autonomous GPS waypoint tracking and real-time telemetry streaming.

AI-Integrated Tonometry System

Patent No. 202511115771

- Designed an ergonomic optical-electronic diagnostic rig, cutting production cost by >80% relative to clinical devices.
- Streamlined patient alignment mechanisms to enable low-cost, portable intraocular screening in underserved areas.

Liquid Metal 3D Printing System

Patent No. 202511034015, 202511099165

- Integrated furnace-nozzle thermal feedback with multi-axis CNC kinematics for precise direct-write deposition of molten alloys.
- Formulated multi-material flow-mixing protocols, enabling continuous composition-graded metal alloy deposition.

Specialized E-Mobility System for Pediatric Dystonia/Cerebral Palsy

Capstone Project

- Led a 6-member team to design a dynamic-backrest e-mobility system for pediatric dystonia/cerebral palsy patients, grounding requirements in a user-centered caregiver survey of 17 patients.
- Designed a tuned spring-damper mechanism absorbing dynamic shock loads up to 200% of patient body weight during acute dystonic events.
- Optimized space-frame tubular truss geometry via FEA, restricting peak dynamic structural deflection to ~0.17 mm.

Self-Balancing Robot

- Implemented a real-time PID feedback control algorithm running at 100 Hz (10 ms sampling rate) on an ESP32 using 6-axis IMU sensor fusion.

LEADERSHIP EXPERIENCE

Society of Intelligent Systems

Feb 2025 – Jun 2026

Project Lead

Patiala, Punjab

- Led 25+ engineers across mechanical, embedded, and software divisions to deliver autonomous robotic platforms.
- Instituted interface design reviews across CAD and ROS nodes, eliminating cross-subsystem integration failures.

TECHNICAL SKILLS

Software: C++, Python, C, Arduino IDE, ROS, MATLAB, Git, Maxsurf, SolidWorks, Linux, AutoCAD, Eagle, Creo

Hardware: Arduino, ESP32, Raspberry Pi, GNSS/GPS Modules, Sensor Integration, Embedded Systems, PCB Prototyping, CAD & Mechanical Design, 3D Printing, Robotics System Integration

AWARDS

Gold Medalist 2025 – Rank #1, Highest CGPA (2nd Year), Mechatronics

ROBO-AI Industrial Training, 2024 (80 hrs) – Robotics, Automation, AI (NVIDIA/MSFT/AWS)